

Learning A Personalized Generative Prior For Face Images

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Inpainting

Super-Resolution (x32)

Semantic Editing

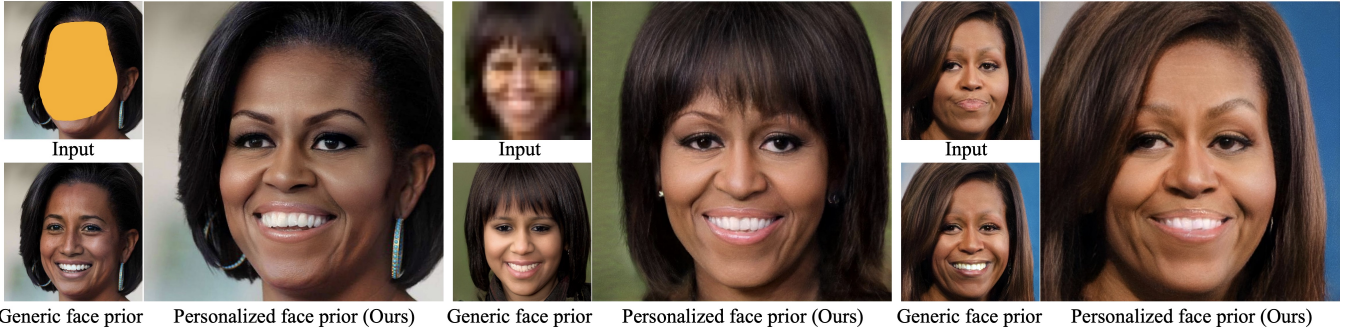


Figure 1: Using our personalized prior for Michelle Obama, we solve inpainting, super-resolution, and semantic editing (smile), while faithfully preserving her key facial characteristic. Each example shows the original input image, which may be corrupted (top left), and the output based on our personalized (right) and generic (bottom left) face priors. The generic face prior is learned from a diverse set of images and produces results that do not preserve Obama’s key facial characteristics. Left and middle blocks ©U.S. Government, right block ©Tim Pierce.

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1 INTRODUCTION

In recent years, the domain of image editing and enhancement in general, and face editing in particular, has experienced a significant shift. From pixel-level approaches, the field gradually shifted to latent-space methods that essentially interpret editing operators applied in a latent space of a generative model, as explicit image

editing operators. This latent-space-based editing enables new capabilities and demonstrates state-of-the-art performance. In particular, StyleGAN [Karras et al. 2020] became the gold-standard and core component for intuitive editing of face images.

While results are impressive, all methods operate by hallucinating information from a general *domain prior* that is learned from a large and diverse dataset containing many identities. Hence, when editing an image of a recognizable person, the result of using a generic face prior may be a person that resembles them but does not preserve their key facial features (see Figure 1).

In this work, we propose an approach to a new problem - *personalization* of a face prior, which addresses the following question: Given a few portrait images (~ 100) of an individual, can we learn a *personalized prior* that facilitates semantic editing and enhancement operations while being faithful to the unique facial characteristics of that individual?

2 METHOD

Adaptation: Given a personalized reference set of N images, $\mathcal{D}_p = \{x_i\}_{i=1}^N$, we adapt a pre-trained StyleGAN generator, G_d , constituting a domain prior into one constituting a personalized prior, G_p . Our adaptation method is inspired by Roich et al. [2021].

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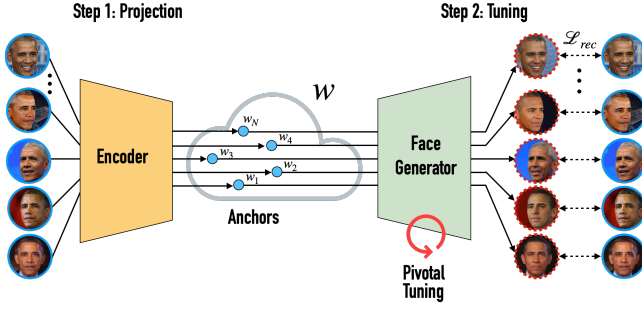


Figure 2: An illustration of our tuning procedure.

We first find the set's nearest-neighbors in G_d by inverting each image into the latent space. This results in a set of latent codes, $\{w_i\}_{i=1}^N$, which we call *anchors*. As $G_d(w_i)$ is still different from x_i , we tune G_d with a simple reconstruction objective – reconstruct x_i given the latent code w_i . Formally,

$$G_p = \arg \min_G [\mathbb{E}_{x_i \in \mathcal{D}_p} \mathcal{L}_{lips}(G(w_i), x_i) + \|G(w_i) - x_i\|_2]. \quad (1)$$

Following the adaptation, the identity of the individual is faithfully depicted in the anchors, but not only there. Close neighborhood of the anchors and interpolations between two anchors have also been personalized. We speculate this behavior is caused by two well-known properties of StyleGAN's latent space – smoothness and disentanglement [Karras et al. 2020]. A different identity in the close neighborhood of an anchor would contradict smoothness, while different identity in the interpolation of two anchors would contradict disentanglement. We speculate that it is easier for the generator to preserve these two properties and thus the training propagates outwards from the anchors.

To capture the personalized regions, we therefore define a new latent space, coined $\mathcal{P}_\beta$, which is the convex hull (captures interpolations) spanned by anchors when dilated outwards (captures neighborhood). β marks the amount of dilation and is often omitted.

$$\mathcal{P}_\beta = \left\{ \sum_i \alpha_i w_i \mid \sum_i \alpha_i = 1, \forall i : \alpha_i \geq -\beta \right\}. \quad (2)$$

Personalized Enhancement and Editing: Having obtained a personalized prior, we leverage it for downstream applications – inpainting, super-resolution and semantic editing. Such applications have already been solved in a non-personalized manner using StyleGAN's domain prior. Our code idea is to carefully adapt existing techniques to use the personalized latent space $\mathcal{P}_\beta$.

For image enhancement, we adopt the projection approach used by Yang et al. [2021]. Their method could be summarized as “find the latent code that after degradation best reconstructs an input image”. We adapt their method to use the $\mathcal{P}$ space. This adaptation is easy to implement in the space of the coefficients of the linear combinations in $\mathcal{P}$, which we call the α -space. We accordingly perform the optimization in that space. (see Figure 3).

For semantic editing, we start from generic, non-personalized, editing directions found in StyleGAN. To edit an image latent space, existing method commonly traverse along a linear direction $\mathbf{n}$ starting from a latent code that corresponds to the image. To make such a direction personalized, we first project it to the span of the anchors. Next, if the direction has been traversed “too far”, it may leave $\mathcal{P}$

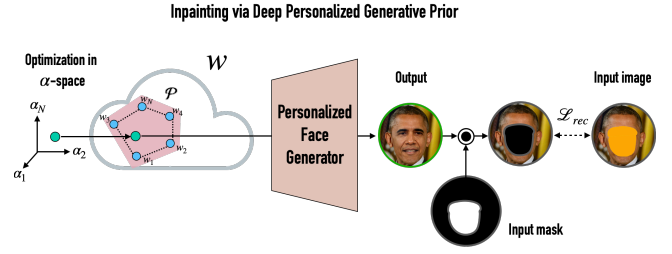
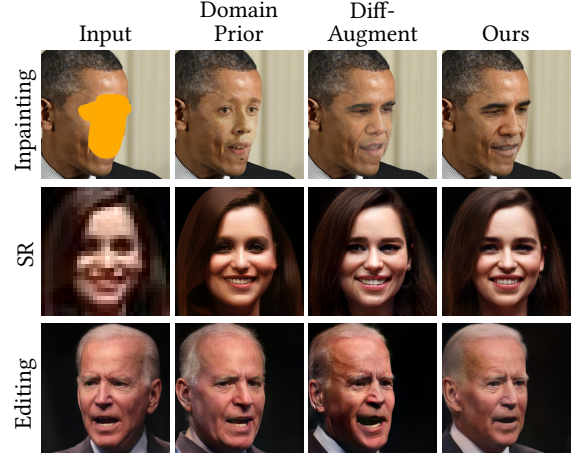
Figure 3: Illustration of our enhancement method for Inpainting. We optimize a latent code in α -space to find the latent code in $\mathcal{P}$ that most accurately reconstruct the corrupted input image, outside of the mask.

Figure 4: Comparison of our personalized enhancement with alternative approaches – domain-prior methods and a prior obtained using a few-shot adaption method [Zhao et al. 2020]. Copyrights from top to bottom ©EJ Hersom, Gage Skidmore, Gage Skidmore.

and thus seize being personalized. To counter this possibility, we project the edited latent back to $\mathcal{P}$.

3 CONCLUSIONS

The results presented in Figure 1 and Figure 4 indicate that the methods introduced are able to faithfully form a personalized prior from a small set of images and leverage it for ill-posed tasks. Interesting directions for future research include incorporating several personalized space in a single generator and accounting for varying appearance of the same individual over time.

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